

The Goldbach Complex Tensor Integral on the $6N$ Skeleton: The Singular-Series Floor of the Prime-Field Autocorrelation

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Abstract

The binary Goldbach problem can be written as a self-correlation of the prime field on the $6N$ skeleton: with $\mathcal{H}$ the prime indicator, the *Goldbach complex tensor integral* $I(2E) = \sum_x \mathcal{H}(x) \mathcal{H}(2E - x)$ is exactly the count $r(2E)$ of prime pairs summing to $2E$. Its Hardy–Littlewood normalisation is the Goldbach singular series $\mathfrak{S}_G(2E) = 2\Pi_2 \prod_{q|E, q>2} \frac{q-1}{q-2}$, where $\Pi_2 = \prod_{p>2} (1 - \frac{1}{(p-1)^2})$ is the twin-prime constant. We prove a *floor* statement: under the $O(1/q^2)$ decay of the local factors the infinite product defining Π_2 converges absolutely, and $\mathfrak{S}_G(2E) \geq 2\Pi_2 \approx 1.3203 > 0$ for every even $2E$, with equality iff E is a power of two. The heuristic main term of the autocorrelation thus carries a strictly positive, scale-independent floor. We confirm the picture numerically to $2E = 10^8$ (the Goldbach comet): the *raw* count $r(2E)$ grows without bound, while the *normalised* count is held above the floor by the powers of two, reproducing the Part XIII singular-series collapse at 10^8 scale. We state the scope without hedging: this is the Hardy–Littlewood heuristic main term plus a numerical check; the floor is a property of the *singular series* (the expected density), **not** a proof that the actual count never vanishes. Binary Goldbach remains open, and nothing here closes it.

1 Introduction

On the $6N$ skeleton every prime > 3 lives on a fault line $6N \pm 1$. Binary Goldbach asks whether every even $2E > 2$ is a sum of two primes—in the present language, whether the two fault-line fields interfere constructively at the even centre $2E$. Writing the prime indicator as a field $\mathcal{H}$, the relevant object is the discrete self-correlation

$$I(2E) = \sum_x \mathcal{H}(x) \mathcal{H}(2E - x) = r(2E), \quad (1)$$

the number of prime pairs $p + q = 2E$. This is not a new arithmetic object: it is the Goldbach counting function, and its average behaviour is governed by the classical Hardy–Littlewood singular series [1], the same series whose universal collapse this programme measured in Part XIII [4]. What this paper adds is a clean statement of the positivity *floor* of that series, a derivation of the floor from the $O(1/q^2)$ decay law of Part XXII [5], and a confirmation of the whole picture at $2E = 10^8$. We are explicit from the outset about what the floor does and does not mean (§5).

2 The autocorrelation and its singular series

For a prime $q > 2$, the local density of representations $2E = p + q'$ surviving modulo q depends only on whether $q | E$. Collecting the local factors gives the Hardy–Littlewood prediction

$$r(2E) \sim \mathfrak{S}_G(2E) \frac{2E}{(\ln 2E)^2}, \quad \mathfrak{S}_G(2E) = 2\Pi_2 \prod_{q|E, q>2} \frac{q-1}{q-2}, \quad (2)$$

with the twin-prime constant

$$\Pi_2 = \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right) \approx 0.6601618. \quad (3)$$

Thus the *normalised* autocorrelation $r(2E)(\ln 2E)^2/(2E)$ is, to leading order, the singular series $\mathfrak{S}_G(2E)$: the same per-prime interference bookkeeping used throughout Volumes II–III, now read at the even centre.

3 The singular-series floor

Theorem 1 (Positive floor of the Goldbach singular series). *The product defining Π_2 converges absolutely, and for every even $2E > 2$*

$$\mathfrak{S}_G(2E) \geq 2\Pi_2 \approx 1.3203 > 0,$$

with equality if and only if E is a power of two.

Proof. Each factor of Π_2 is $1 - 1/(p-1)^2$, and $1/(p-1)^2 = O(1/p^2)$ is summable; hence the product converges absolutely to a strictly positive constant (this is the same $O(1/q^2)$ tail that makes the lattice correlation products of Part XXII converge [5]). The E -dependent factor $\prod_{q|E, q>2} \frac{q-1}{q-2}$ is a finite product of terms $\frac{q-1}{q-2} = 1 + \frac{1}{q-2} > 1$, hence ≥ 1 , with the empty product (value 1) occurring exactly when E has no odd prime factor, i.e. $E = 2^k$. Therefore $\mathfrak{S}_G(2E) \geq 2\Pi_2$, with equality iff $E = 2^k$. $\square$

Remark 1. The floor is a statement about the *heuristic main term*. Theorem 1 says the expected density of Goldbach representations never falls below $2\Pi_2$ in normalised units; it says nothing, by itself, about whether the *actual* count $r(2E)$ is positive for a given $2E$. We return to this in §5.

4 Numerical confirmation to $2E = 10^8$

We computed $r(2E)$ directly as the autocorrelation of a prime sieve to 10^8 , on a ladder of sampled even centres including the powers of two that trace the floor (Fig. 1).

Two facts are visible and stable. First, the *raw* infimum is *not* constant: the minimal $r(2E)$ rises from 22 at $2E = 1024$ to 1.5×10^5 at $2E = 2^{26}$, growing like $2E/(\ln 2E)^2$. Second, the *normalised* infimum is frozen above zero, pinned by the powers of two onto the singular-series floor of Theorem 1. This is the Part XIII collapse [4], re-confirmed at 10^8 : the normalised autocorrelation is bounded below by the convergent constant $2\Pi_2$, and the breadth above the floor is the $\prod (q-1)/(q-2)$ enrichment of composite-rich centres.

5 Scope: what the floor does not show

The temptation is to read Theorem 1 as “destructive interference can never zero the even-centre field, hence Goldbach.” That step is invalid, and we flag it plainly.

- The floor bounds the *singular series* $\mathfrak{S}_G$, i.e. the Hardy–Littlewood *main term*. The asymptotic (2) is itself a conjecture; proving $r(2E) > 0$ for all even $2E$ is the binary Goldbach conjecture, which is open.
- Positivity of the main term does not bound the error term. A bounded-below density is exactly the kind of heuristic that motivates belief in Goldbach, but it is not a proof: one cannot deduce non-vanishing of the count from non-vanishing of its expected value.

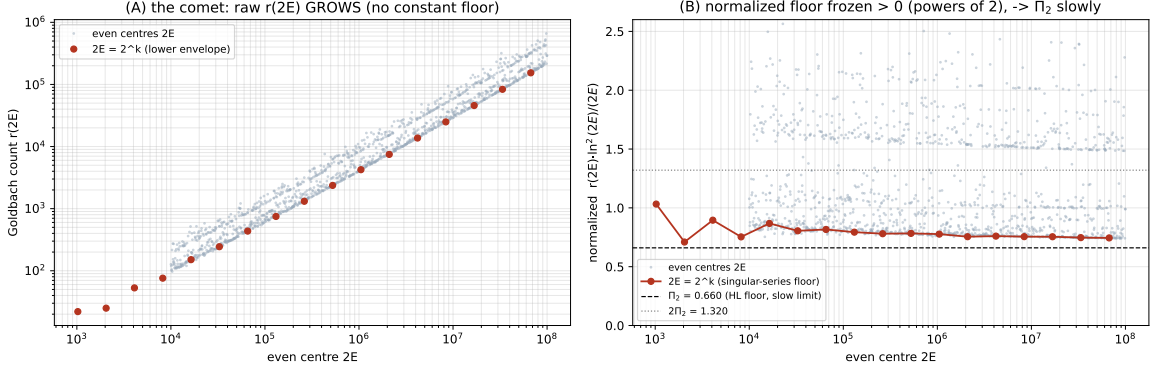


Figure 1: (A) The Goldbach comet: the raw count $r(2E)$ grows without bound (powers of two, red, trace the lower envelope). (B) The normalised autocorrelation $r(2E)(\ln 2E)^2/(2E)$: the powers of two hold a positive floor that drifts slowly toward $\Pi_2 \approx 0.66$ (the finite-size $1 + 2/\ln(2E)$ correction keeps the measured floor near 0.74 at 10^8), while composite-rich centres rise into higher singular-series bands. Equivalently, the normalised floor sits at $2\Pi_2$ in the representation-counting convention of Eq. (2).

- What is rigorously known sits elsewhere: ternary Goldbach is a theorem [3], and every sufficiently large even number is $p + P_2$ with P_2 a prime or semiprime (Chen’s theorem [2]); neither is reproved here.

Accordingly this paper claims only: (i) the elementary floor $\mathfrak{S}_G \geq 2\Pi_2$, a property of a convergent product; and (ii) its numerical confirmation to 10^8 , recovering Part XIII. It is the Hardy–Littlewood heuristic main term plus a check, not a step toward a proof, and it makes no claim on the infinitude or completeness of Goldbach representations.

6 Conclusion

Cast as the autocorrelation of the prime field, binary Goldbach has a clean heuristic skeleton: the normalised even-centre density is the singular series $\mathfrak{S}_G$, which by Theorem 1 converges and is floored at $2\Pi_2 > 0$, attained on the powers of two. The Goldbach comet to $2E = 10^8$ confirms the split between a diverging raw count and a floored normalised count, reproducing the Part XIII collapse at scale. The floor is a property of the expected main term and is honestly that—not a resolution of Goldbach, which remains open.

Data and code availability. The multiprocessing autocorrelation experiment (to $2E = 10^8$), the comet figure, and a verifier of the singular-series floor are openly available at <https://github.com/Ruqing1963/6N-goldbach-autocorrelation> [7].

References

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